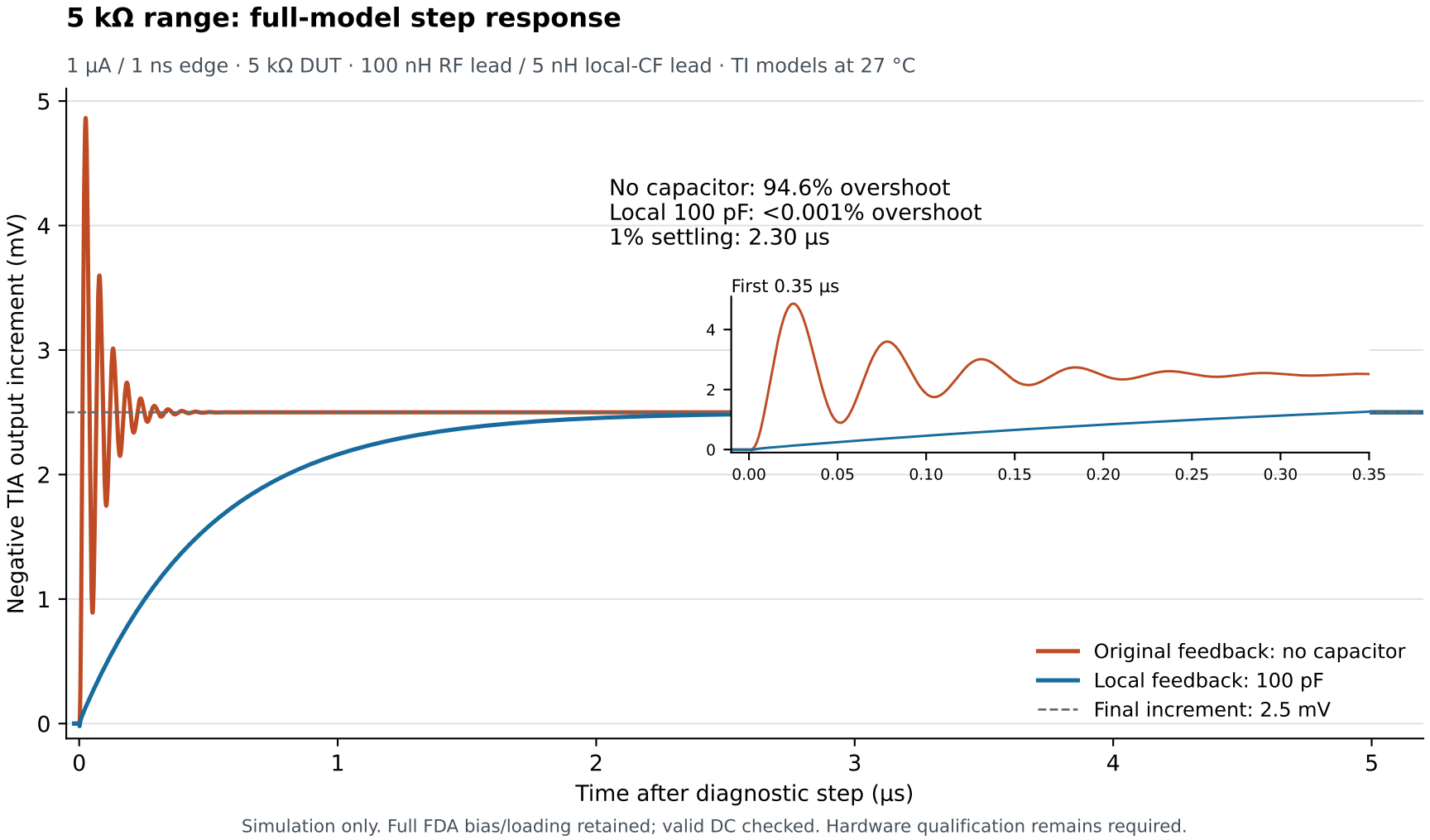
5 kΩ range: full-model step response
1 µA / 1 ns edge · 5 kΩ DUT · 100 nH RF lead / 5 nH local-CF lead · TI models at 27 °C
No capacitor: 94.6% overshoot
Local 100 pF: <0.001% overshoot
1% settling: 2.30 µs
First 0.35 µs
Negative TIA output increment (mV)
Time after diagnostic step (µs)
Original feedback: no capacitor
Local feedback: 100 pF
Final increment: 2.5 mV
Simulation only. Full FDA bias/loading retained; valid DC checked. Hardware qualification remains required.